13. A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?

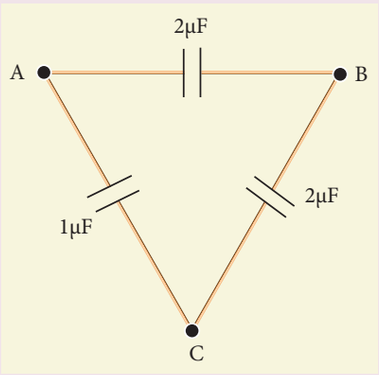
(a) Capacitance

(b) Charge

(c) Voltage

(d) Energy density

14. Three capacitors are connected in triangle as shown in the figure. The equivalent capacitance between the points A and C is



(a) 1μF (b) 2 μF

(c) 3 μF (d) $\frac{1}{4}$μF

15. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is (AIIPMT -2012)

(a) 3×10^{-2} C

(b) 4×10^{-2} C

(c) 1×10^{-2} C

(d) 2×10^{-2} C

Answers

1) b	**2)** c	**3)** d	**4)** b	**5)** a
6) b	**7)** c	**8)** a	**9)** c	**10)** b
11) a	**12)** c	**13)** d	**14)** b	**15)** a

II Short Answer Questions

1. What is meant by quantisation of charges?

2. Write down Coulomb's law in vector form and mention what each term represents.

3. What are the differences between Coulomb force and gravitational force?

4. Write a short note on superposition principle.

5. Define 'electric field'.

6. What is meant by 'electric field lines'?

7. The electric field lines never intersect. Justify.

8. Define 'electric dipole'. Give the expression for the magnitiude of its electric dipole moment and the direction.

9. Write the general definition of electric dipole moment for a collection of point charge.

10. Define 'electrostatic potential".

11. What is an equipotential surface?

12. What are the properties of an equipotential surface?

13. Give the relation between electric field and electric potential.

14. Define 'electrostatic potential energy'.

15. Define 'electric flux'.

16. What is meant by electrostatic energy density?

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